Hand Gesture Recognition: Upgrading the A1 Baseline with DINOv2, Synthetic Stress Conditions and an Interactive Demo

Assignment 2 : Level-Up A1 Topic with SOTA CV + Demo
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Summary. This report continues Assignment 1 (recognising three hand gestures: Peace, Stop, Thumbs Up) on the same self-collected dataset of 372 photos. The A1 baseline CNN reached 83.93% accuracy on the original test split, but it had never been tested on conditions that differ from my training photos. In A2 I (1) add a modern SOTA component, **DINOv2** (a self-supervised Vision Transformer), as a frozen feature extractor with a logistic-regression classifier; (2) collect a new real-world **stress-test set of 96 photos** (dark, backlight, motion blur, new backgrounds); (3) generate **synthetic** difficult conditions programmatically for training and run an ablation; (4) add two explanation methods (attention rollout and occlusion sensitivity); and (5) build an interactive **Gradio demo** that reports latency.

Key results. On the original 56-image test set DINOv2 reaches 100% accuracy (A1 CNN: 83.93%), but this test is small and very similar to the training photos, so I do not treat it as proof. On the stress test the A1 CNN reaches 35.4%, DINOv2 42.7%, and DINOv2 trained with synthetic conditions 49.0% (chance level is 33%). The main finding is therefore not the high in-distribution score but how much robustness is still missing.

1. Introduction

Hand gesture recognition lets a system understand a hand sign from an image, which is useful for touchless interfaces and accessibility. In Assignment 1 I trained a simple CNN (83.93% test accuracy) and a fine-tuned MobileNetV2 (80.36%) on 372 photos. Both were evaluated on a 56-image test split taken from the same photo sessions as the training data. Real use is different: lighting changes, the hand moves, and the background is new. Assignment 2 therefore asks two questions: **(a)** does a stronger pretrained representation (SOTA) improve on the A1 baseline, and **(b)** does the result survive realistic condition changes? I keep the same topic, classes, split and random seed (42) so that the comparison with A1 is fair.

2. Dataset Update

Subset	Images	Peace	Stop	Thumbs Up	How it is used
Original photos (A1, unchanged)	372	127	121	124	Split 260 train / 56 validation / 56 test (stratified, seed 42)
Synthetic images (new)	1,040	-	-	-	Training only: 4 variants generated from each of the 260 training photos
Stress-test photos (new, real)	96	31	33	32	Evaluation only: never used for training or model selection

Table 1. Data used in A2. Classes are approximately balanced in both the original set and the stress test.

Size, balance and synthetic share. The original dataset is unchanged (372 photos, 121 to 127 per class). The final training set has 1,300 images, of which 1,040 (**80%**) are synthetic. Validation and test sets contain no synthetic images. The stress-test photos were taken with a phone after the A1 dataset was finalised, in four conditions: backlight (23 images), blur (25), dark (22) and new background (26). Each condition has 6 to 10 images per class. The A1 augmentation (horizontal flip, small rotations, zoom, translation and contrast changes) is unchanged and was used only when training the A1 CNN and MobileNetV2; DINOv2 features are extracted from the original, un-augmented training photos plus the synthetic variants described below.

Synthetic condition generator. For each training photo I create four variants. Each variant gets a random crop (55 to 100% of the area), a horizontal flip with probability 0.5, and one of four conditions chosen at random: *dark* (brightness x0.2 to x0.6), *backlight* (brightness x1.4 to x2.0 with lower contrast), *motion blur* (box blur of 5 to 14 pixels along a random axis) or *noise* (Gaussian, sigma 10 to 35 grey levels). These are simple image operations, not photorealistic generation. All synthetic images are generated offline from the training split only and are labelled SYNTHETIC in figures and in the app. The heavy generators recommended in the assignment (for example Step-Video-T2V with about 30 billion parameters) do not fit the free Colab GPU, so I use this lightweight substitute and state it explicitly.



Figure 1. Examples of synthetic training images (all marked SYNTHETIC). They are generated from training photos only.

3. SOTA Method: DINOv2

What it is. DINOv2 (Meta AI, 2023) is a family of Vision Transformers trained with self-supervised learning on a large curated image collection (about 142 million images) without any labels [1]. A Vision Transformer [5] cuts the image into small patches (here 14 x 14 pixels, so a 224 x 224 image gives 16 x 16 = 256 patches), turns each patch into a vector (a token), and lets all tokens exchange information through self-attention layers. An additional CLS token summarises the whole image. DINOv2 is trained so that different crops and distortions of the same image get consistent features, which makes the features describe shape and layout well. The authors show that a simple linear classifier on top of the frozen features already works well (linear probing) [1]. I use the smallest model, *dinov2-small* (about 21 million parameters), which runs on the free Colab GPU in real time.

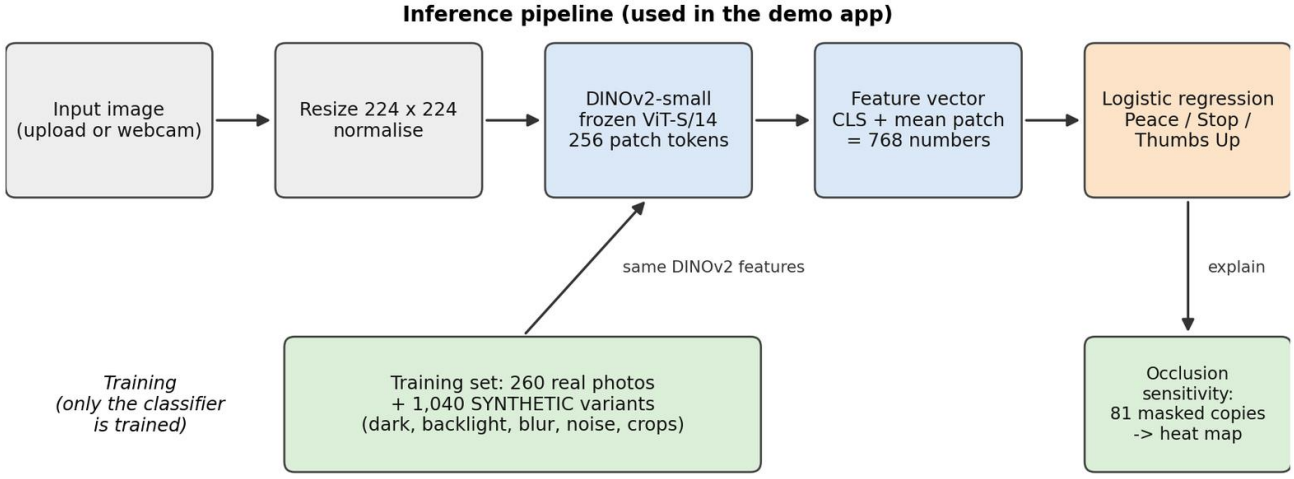


Figure 2. Pipeline. Only the logistic-regression classifier is trained; DINOv2 stays frozen.

My implementation. Each image is resized to 224 x 224, normalised with ImageNet statistics and passed through the frozen encoder. The CLS token and the mean of the patch tokens are concatenated into a 768-dimensional vector (2 x 384). A logistic-regression classifier (L2 regularisation, C = 1) is trained on these vectors. Training takes seconds.

Why it should help this topic.

- **Small data.** With 260 training photos, a CNN trained from scratch (A1) has to learn visual structure from very little data. Frozen features trained on about 142 million images already encode shape and texture.
- **Appearance changes.** Features trained to stay consistent under crops, colour changes and blur should be less sensitive to the conditions measured by my stress test.
- **Cheap iteration.** Only the small classifier is retrained, so each experiment (for example adding synthetic data) takes seconds instead of a full network training.

Scope decision. The assignment table lists Step-Video-T2V and ArtAug for gesture topics but also allows choosing another modern SOTA. I chose DINOv2 because it can be run and evaluated completely in free Colab within the deadline, and because it addresses the robustness question directly. I cite the original papers [1, 2] instead of a Neurohive page.

4. Experiments

Protocol. Same stratified 70/15/15 split and seed 42 as in A1 (260 / 56 / 56). A1 numbers come from the A1 notebook. The stress test is used for evaluation only; no hyper-parameter was tuned on it. I report accuracy and macro F1 on the original test set and accuracy per condition on the stress test (macro F1 is not reported per condition because each condition has only 6 to 10 images per class).

4.1 Original test set (56 images)

Model	Accuracy	Macro precision	Macro recall	Macro F1
A1 Baseline CNN	83.93	83.92	84.02	83.93
A1 MobileNetV2 (frozen)	48.21	-	-	-
A1 MobileNetV2 (fine-tuned)	80.36	82.97	80.70	79.56
DINOv2 + LogReg (real data only)	100.00	100.00	100.00	100.00
DINOv2 + LogReg (+ synthetic)	100.00	100.00	100.00	100.00

Table 2. Original test set results (%). A1 values are taken from the A1 notebook.

DINOv2 classifies all 56 test images correctly (validation accuracy is also 100%). However, one image is worth 1.8 percentage points in this test set, and the learning curve (Figure 5) shows validation accuracy reaching 100% with only 156 training images. A ceiling like this means the test can no longer separate a good model from an excellent one. Photos were also taken in bursts, so near-duplicate images may exist across splits, which would inflate in-distribution accuracy. I therefore do not read 100% as “solved” and rely on the stress test below.

4.2 Stress test (96 new photos) and ablation

Model	Backlight (23)	Blur (25)	Dark (22)	New background (26)	Overall (96)
A1 Baseline CNN	39.1 (9)	28.0 (7)	27.3 (6)	46.2 (12)	35.4 (34)
DINOv2 + LogReg (real only)	39.1 (9)	36.0 (9)	40.9 (9)	53.8 (14)	42.7 (41)
DINOv2 + LogReg (+ synthetic)	39.1 (9)	36.0 (9)	54.5 (12)	65.4 (17)	49.0 (47)

Table 3. Stress-test accuracy in % (number of correctly classified images in brackets). Chance level is 33%. The last row is the ablation: the same model trained with and without synthetic conditions.

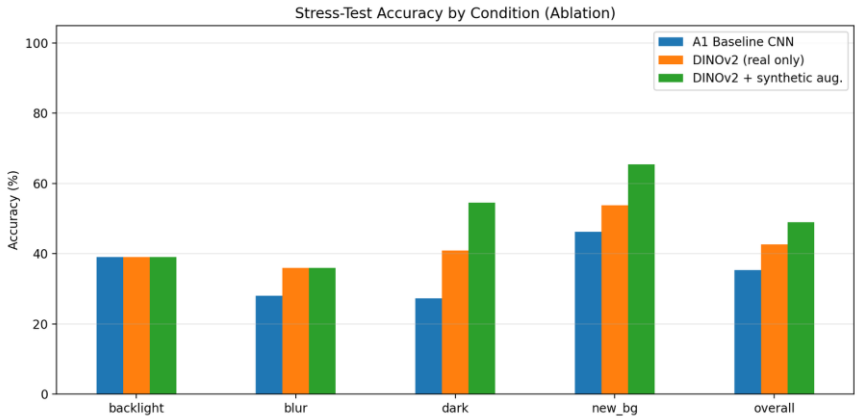


Figure 3. Stress-test accuracy by condition: A1 baseline vs DINOv2 vs DINOv2 trained with synthetic conditions (ablation).

DINOv2 vs A1. DINOv2 is better than the A1 CNN in three of four conditions (tied on backlight) and +7.3 percentage points overall. But 42.7% is only slightly above chance. The confusion matrix of the real-only model shows why: it predicts “stop” for 69 of the 96 stress images. Per-class recall is 4/31 for Peace (13%), 26/33 for Stop (79%) and 11/32 for Thumbs Up (34%), so in unfamiliar scenes the model collapses towards “stop”. Some dark photos are almost black and the gesture is invisible even to a human, so part of the error cannot be fixed by the model.

Ablation: synthetic conditions. Training with the 1,040 synthetic images (80% of the training set) raised stress accuracy from 42.7% to 49.0% (6 more images correct), while the original test accuracy stayed at 100%. The gain comes from *dark* (+3 images, 40.9% to 54.5%) and *new background* (+3 images, 53.8% to 65.4%); the latter probably because random crops and flips change the composition. *Backlight* and *blur* did not change at all, which suggests that my synthetic backlight and blur do not resemble the real ones (the real backlight photos contain glare and floor reflections that a brightness and contrast change does not reproduce). **Caution:** with 96 images the 95% confidence interval of an accuracy near 45% is roughly +/-10 percentage points, and this is a single run, so differences of 3 to 7 images are suggestive, not conclusive.

4.3 Training curves

The A1 models were trained for epochs, so their curves are reproduced from the A1 notebook (Figure 4). The baseline CNN's validation accuracy peaks around epoch 15 and then becomes unstable, which is typical for a small validation set of 56 images. DINOv2 + logistic regression has no epochs, so I show a **learning curve** instead (Figure 5): the classifier is trained on 10% to 100% of the real training features.

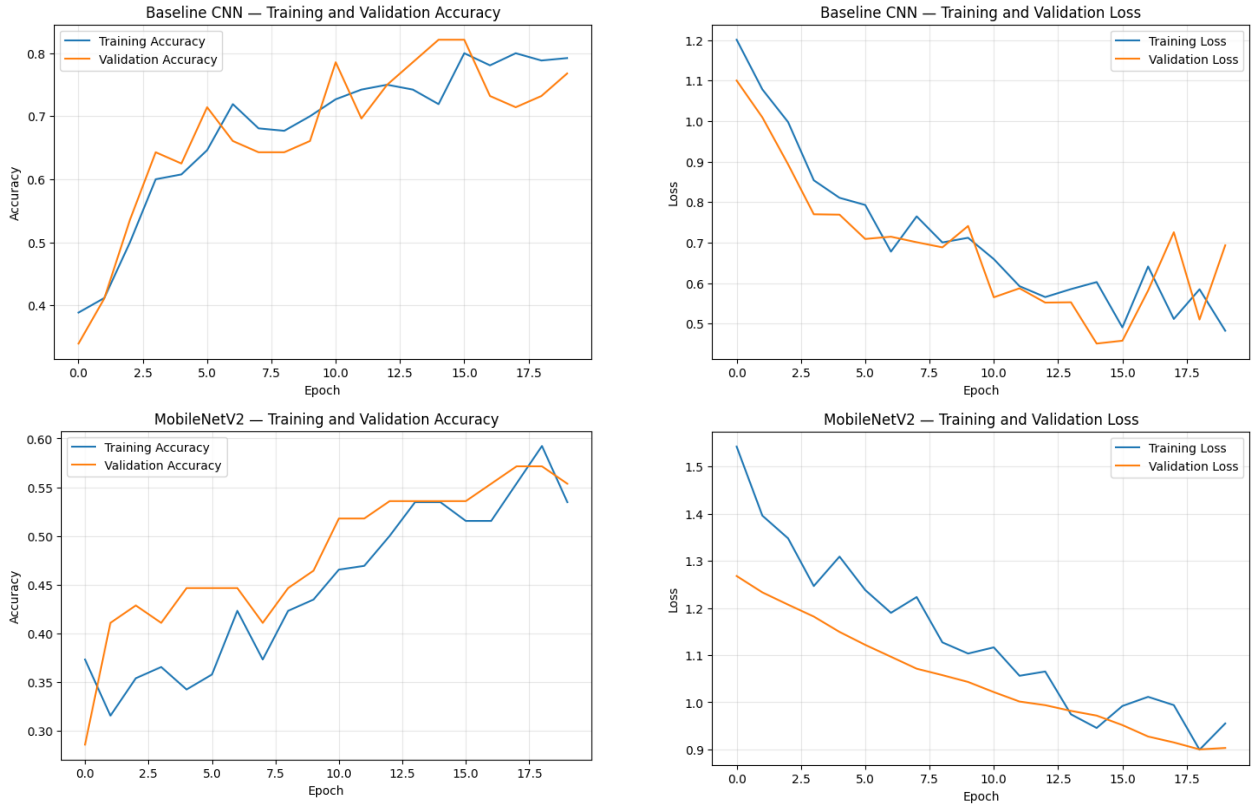
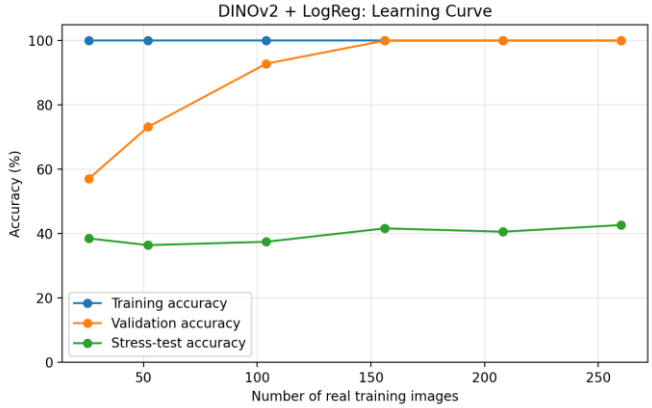


Figure 4. A1 training and validation curves (from the A1 notebook): baseline CNN (top) and MobileNetV2 with frozen base (bottom).



Reading the learning curve. Training accuracy is 100% at every size. Validation accuracy rises from 57% (26 images) to 100% (156 images). Stress-test accuracy, however, stays between 36% and 43% regardless of how many training photos are used. This indicates that the problem is not the number of photos but their **diversity**: more photos of the same kind do not teach the model new conditions. The stress-test numbers here are for evaluation only and were not used to choose anything.

Figure 5. Learning curve of DINOv2 + logistic regression (real training images only).

5. Explainability

I tried two methods on the final classifier. **Attention rollout** [4] multiplies the self-attention matrices of all transformer layers to show where the CLS token attends. **Occlusion sensitivity** [3] slides a grey 48 x 48 square over the image (stride 24, 81 positions) and measures how much the predicted-class probability drops; red marks regions the prediction depends on. Occlusion needs only the model's predictions, so it measures real dependence of the output on image regions.

Observations (Figure 6). In several cases the red region is on the hand (test Stop, backlight Thumbs Up, backlight Stop, dark Stop). In others it lies on the background or is spread over the whole frame (test Peace example 2, dark Thumbs Up, blur on the bedspread), so the model uses context as well as the hand, which is consistent with its weak results on new backgrounds. Several wrong predictions have very high confidence (for example backlight Thumbs Up predicted as Stop with 100%), so the model's confidence cannot be trusted as a measure of correctness.

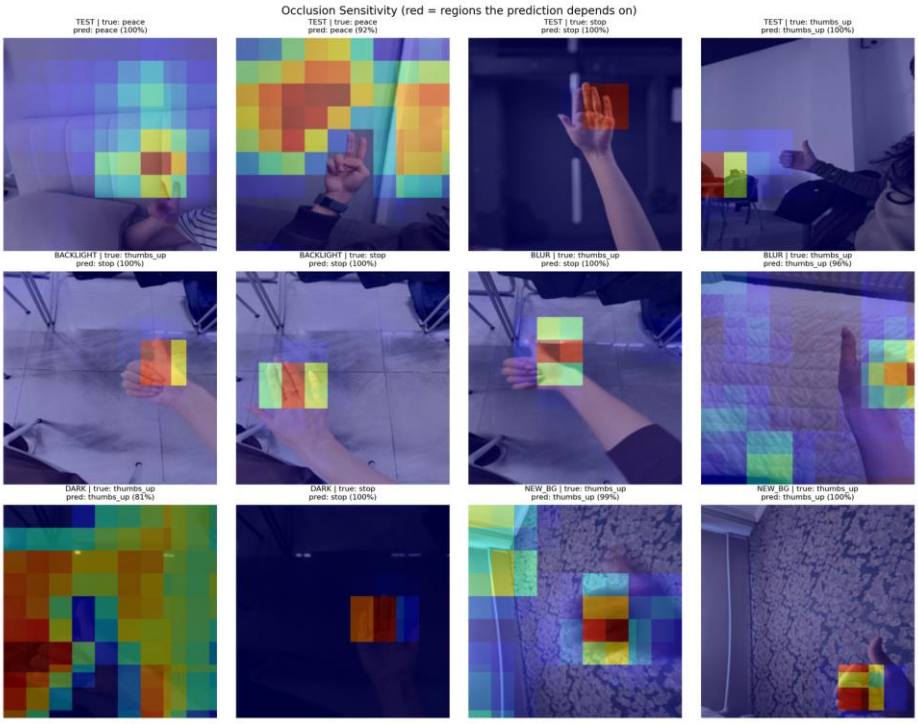
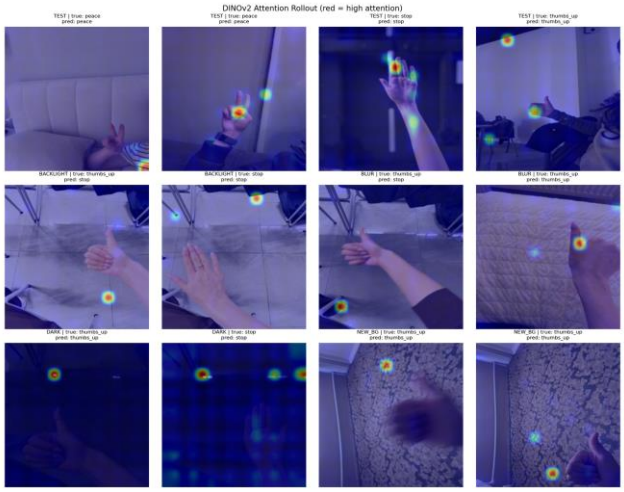


Figure 6. Occlusion sensitivity on test images (first row) and stress-test images. Titles show the true class, the prediction and its confidence.



Attention rollout was not reliable here. The maps (Figure 7) show small isolated bright spots that are often not on the hand, for example on lamps and reflections in dark images or on the floor. Vision Transformers without register tokens are known to create such high-attention artifact tokens in uninformative areas [2], so I do not use rollout for conclusions and keep occlusion as the main explanation. Reporting this negative result is deliberate: a map that looks attractive is not necessarily faithful.

Figure 7. Attention rollout on the same images (red = high attention).

6. Interactive Demo Application

The demo is a Gradio web app (file *app.py* in the repository). The flow is: **upload or webcam photo, prediction, visual explanation**. A stress-test generator (dropdown: Dark, Backlight, Motion blur, Noise, with a strength slider) modifies the input; the modified image is shown with the label SYNTHETIC. The app shows class probabilities, the model input, the occlusion-sensitivity map, warnings (very dark image, low confidence), the inference time and the device. A public link was created from Google Colab (T4 GPU); the same code runs locally with *python app.py*.

Measurement (Colab, Tesla T4 GPU)	Time per image	What is included
A1 baseline CNN (Keras)	14.7 ms	Model forward pass, image already resized in memory
DINOv2 + LogReg (offline measurement)	49.8 ms	Reading the file, preprocessing, DINOv2, classifier
DINOv2 + LogReg (inside the app)	24 ms	Preprocessing from a PIL image in memory, DINOv2, classifier
Occlusion explanation	689 ms	81 masked copies through DINOv2 + classifier

Table 4. Latency. The measurements have different scopes, so they are not directly comparable; single measurements, no repeated-run statistics.

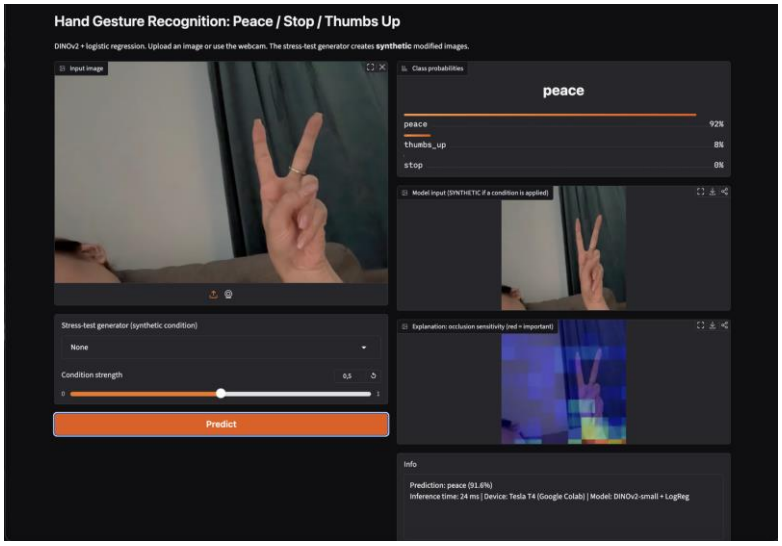


Figure 8. Screenshot of the running demo (Colab public link, webcam/upload input, probabilities, model input, occlusion map, latency and device).

7. Ethics and Limitations

- **Data and consent.** All photos (original and stress test) were taken by me and show my own hands; some backgrounds partly show other people, and the raw dataset is kept private and is not included in the repository. The app code does not intentionally save uploaded images.
- **Synthetic media.** Synthetic images are simple programmatic edits of my own photos, not generated people or deepfakes. They are used for training only, never in test sets, and are labelled SYNTHETIC in figures and in the app. No generated faces or avatars are used.
- **Bias.** The images come mostly from one person's hands and a small number of places. Performance for other hands (skin tone, size, age, left or right hand), cameras and backgrounds is unknown and is expected to be worse. Do not use this model for decisions about people.
- **Misuse and reliability.** Gesture control is low risk, but wrong confident predictions (Section 5) mean it must not be used in safety-critical settings. The app warns about very dark images and low confidence.
- **Limitations.** Small dataset (372 photos, 3 classes); stress test collected by one person with 96 images, so results have wide confidence intervals; single runs without repeated seeds; the 100% in-distribution score is probably optimistic; synthetic backlight and blur did not help; frozen small DINOv2 was not fine-tuned; the recommended generators (Step-Video-T2V, ArtAug) were not used.

8. Reproducibility

Environment. Google Colab (free), Tesla T4 GPU, Python 3; libraries: transformers, torch, scikit-learn, pillow, pillow-heif, matplotlib, pandas, tensorflow (only to load the A1 CNN), gradio. Random seed 42; the split files *split_train.csv*, *split_val.csv* and *split_test.csv* are saved.

Weights. DINOv2 weights are downloaded automatically from Hugging Face (*facebook/dinov2-small*). The trained classifiers are in *dinov2_logreg_synth.joblib* (final) and *dinov2_logreg_real_only.joblib* (ablation), with *class_names.json*. The A1 CNN is *best_hand_gesture_cnn.keras*.

How to run. (1) Notebook: open the A2 notebook in Colab, enable a T4 GPU, mount Drive with the dataset in *Computer_Vision_Assignment/hand_gesture_dataset* and the stress photos in *stress_test/<condition>/<class>*, then run the cells in order. (2) App: *pip install -r requirements.txt*, put the two weight files next to *app.py*, run *python app.py* and open the printed address. The repository link and the demo video link are provided in the Moodle submission.

References

[1] M. Oquab et al. DINOv2: Learning Robust Visual Features without Supervision. arXiv:2304.07193, 2023.

[2] T. Darcet, M. Oquab, J. Mairal, P. Bojanowski. Vision Transformers Need Registers. arXiv:2309.16588, 2023.

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[5] A. Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR, 2021.

[6] M. Sandler et al. MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR, 2018 (A1 transfer-learning model).